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SUPERSEDING
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MILITARY SPECIFICATION

HEATERS, HOT OIL

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes of automatic hot-oil heaters. The heaters shall be skid mounted or trailer mounted, light fuel oil fired, and electric driven.

1.2 Classification. The heaters shall be of the following types and sizes, as specified (see 6.2):

Type I - Skid-mounted.
  Size 1 - 1,500,000 Btuh output.
  Size 2 - 2,100,000 Btuh output.
  Size 3 - 3,000,000 Btuh output.
Type II - Trailer-mounted.
  Size 2 - 2,100,000 Btuh output.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-1106	- Extinguisher, Fire, Carbon Dioxide.
W-C-586	- Conduit Outlets and Entrance Caps, Electrical; Cast Metal - For Shore Use.
FF-B-171	- Bearings, Ball Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearings, Roller, Tapered.
HH-B-671	- Bricks, Refractory: Fire Clay.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
WW-C-566	- Conduit, Metal, Flexible.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-717	- Castable Mix, Refractory, High-Temperature Hydraulic-Setting.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.

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MIL-P-14105	- Paint, Heat-Resisting (for Steel Surfaces).
MIL-H-18160	- Hose Assemblies, Corrugated Steel, 200 Psi and 425° F. Maximum Steam Service Rating.
MIL-I-22023	- Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.
MIL-L-22851	- Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant).
MIL-G-25013	- Grease, Aircraft, Ball and Roller Bearing.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52286	- Cable Assemblies, Power, Electrical, (With Molded on Terminations).
MIL-C-52286/11	- Cable Assembly, Power Electrical.
MIL-C-52286/24	- Cable Assembly, Power Electrical, Adapter.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors.

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DOD-STD-100	- Engineering Drawing Practices.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-642	- Identification Marking of Combat, and Tactical Transport Vehicles.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

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HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for use in Ground Equipment Systems.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Vehicle Safety Standard (FMVSS).
Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME - B31.1 Power Piping.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN TRUCKING ASSOCIATIONS, INC. (ATA)

Regular Common Carrier Conference (RCCC).
RCCC RP#S110 - Vehicle Running Lamps.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, NW, Washington, DC 20036.)

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Welding Procedures and Performance Qualifications.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Northwest Le Jeune Road, Miami, FL 33135.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book.

(Application for copies may be obtained from brake lining supplier or should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Central and Systems.
MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC. (T&RA)

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

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TRUCK-TRAILER MANUFACTURES ASSOCIATION, INC.

TTMA RP#6
TTMA RP#9
TTMA RP#12
TTMA RP#14
TTMA RP#40
TTMA Bulletin No. 65

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., Suite 203, 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

UNDERWRITERS LABORATORIES (UL)

UL 6 - Rigid Metal Coduit.
UL 516B - Fittings for Conduit and Outlet Boxes.
UL 1242 - Immediate Metal Conduit.

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingster Road, Northbrook, IL 60002.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Each heater shall consist of an insulated heat exchanger, an oil burner system, an expansion and cold-oil seal tank, a transfer-oil circulating system, controls and safety devices, and all accessories required for operation (except for burner fuel tank). Heaters shall be ready for use after service connections have been made to a user system.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of heaters from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish heaters conforming to this specification.

3.3 Production heater. All heaters furnished under this specification shall be physically and mechanically identical to the first article heater approved by the Government (see 6.3).

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3.4 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. Heaters shall comply with all applicable FMCSR and FMVSS.

3.5 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.5.1 Material deterioration and control. The heater shall be fabricated from compatible materials that are inherently corrosion resistant or treated in order to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation or storage environment to which the item may be exposed.

3.5.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.5.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or the designated representative (see 6.2).

3.5.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the heater may be newly fabricated from recovered materials to the maximum extent practicable, provided the heater produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the heater.

3.5.1.4 Transfer oil. Unless otherwise specified (see 6.2), transfer oil shall be furnished with each heater. The transfer oil shall conform to MIL-L-22851, type III.

3.5.1.5 Bearings. Unless otherwise specified herein, bearings shall be of the greasable, antifriction type conforming to FF-B-171, FF-B-185, or FF-B-187, as applicable, and shall be of the contractor's recommended capacity for the application. Bearings that are subject to heating by hot transfer oil shall be heat stabilized for operation at a temperature of not less than 475 °F and shall be lubricated with grease conforming to MIL-G-25013.

3.5.1.6 Gaskets and seals. Gaskets and seals that are subject to heating by hot transfer oil shall be oilproofed and shall withstand all temperatures up to 600 °F. Any elastomeric gaskets or seals exposed to hot transfer oil, shall be fabricated from viton or equipment heat-and-oil resistant fluoroelastomer.

3.5.1.7 Refractory. Refractory material shall conform to MIL-C-717 or HH-B-671, as applicable.

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3.5.1.8 Structural steel. Structural steel shall conform to ASTM A 36.

3.5.1.9 Sheet steel. Unless otherwise specified herein, sheet steel shall conform to SAE 1015, 1020, or 1025, hot-rolled, condition and finish to suit the application.

3.5.1.10 Sheet aluminum. Unless otherwise specified herein, sheet aluminum shall conform to QQ-A-250/2, temper H14 or H24.

3.5.1.11 Pipe and tubing. Unless otherwise specified (see 6.2), pipe and tubing shall be seamless steel conforming to ANSI/ASME B31.1 for 150 psi, 475 °F hot-oil service.

3.5.1.12 Valves and pipe fittings. Unless otherwise specified (see 6.2), all valves, caps, couplings, flanges, crosses, elbows, reducers, bends, tees, and nipples shall be made of steel for 150 psi, 475 °F, hot-oil service, or of malleable iron for 300 psi, 475 °F, hot-oil service, as applicable, in accordance with ANSI/ASME B31.1. Fittings shall be provided with threaded or flanged connections and shall withstand without leakage, permanent deformation, or damage 150 psi, 475 °F, hot-oil service when connected. Threaded fittings shall be National taper pipe thread (NPT). Each valve shall be identified to show the appropriate function.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part number.

3.6.1 Common parts. Common hardware and fastening devices, such as bolts, nuts, screws, and similar items, shall be held to a minimum number of types and sizes and shall be threaded to the standards of FED-STD-H28.

3.7 Human factors engineering. The heater shall conform to Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.7 (Ground Workspace Design Requirements), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.8 Performance. The heater, burner, heat exchanger, and main circulating pump shall be capable of performing as follows:

a. The maximum heater output shall be not less than 1,500,000 Btuh for size 1, not less than 2,100,000 Btuh for size 2, and not less than 3,000,000 Btuh for size 3.
b. The efficiency of the heat exchanger shall be such that at its rated output the fuel consumption rate of the burner shall be not more than 16 gallons per hour (gph) for size 1, not more than 26 gph for size 2, and not more than 32 gph for size 3.

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c. The main circulating pump shall have a capacity of not less than 70 gallons per minute (gpm) for size 1 heater, not less than 100 gpm for size 2 heater, and not less than 140 gpm for size 3 heater of the specified transfer oil (see 3.5.1.4) at an outlet temperature of 450 °F and a pump discharge pressure of not less than 50 pounds per square inch gage (psig).

d. The heat exchanger shall maintain a flow rate of not less than 70 gpm for size 1 heater, not less than 100 gpm for size 2 heater, and not less than 140 gpm for size 3 heater of the specified transfer oil (see 3.5.1.4) at an outlet temperature of 450 °F +10 °F -0 °F when the inlet temperature of the transfer oil is not more than 360 °F. The pressure drop of transfer oil between inlet and outlet of heat exchanger shall be not more than 15 psig at 70 gpm for size 1 heater, 100 gpm for size 2 heater, and 140 gpm for size 3 heater.

3.9 Circulating pump motor. Unless otherwise specified herein, the motor shall be 3-phase, 60-cycle, ball-bearing, squirrel-cage-induction class, continuous rating, totally enclosed, fan-cooled, design B, in accordance with NEMA Standard MG1, integral horsepower, class F insulation, temperature rise 95 °C. When specified (see 6.2), the motor shall be design C in accordance with NEMA Standard MG1, integral horsepower, class B insulation, temperature rise 40 °C.

3.9.1 Motor starter. The circulating pump motor shall be provided with a motor starter. The motor starter shall conform to MIL-S-12514, type I, class 2. The starter enclosure shall be in accordance with NEMA Standard ICS 1-110, table 1-110-2, type 4X.

3.9.2 Internal wiring. Internal wiring shall be installed in conduit conforming to WW-C-566 , UL6, UL 514B, or UL 1242, as applicable. Conduit outlets shall conform to W-C-586.

3.9.3 Ground rods. Two ground-rod assemblies for grounding the main power supply shall be provided. Each assembly shall consist of a driving bolt, three 3-foot sections of 5/8-inch-diameter steel rod, and three threaded couplings. The rods shall consist of a steel core with a copper surface bonded to the core by fusion or electrolytic means. Each end of each rod shall be threaded and one end of one rod section shall be provided with a cone point. Figure 1 shows the cone-pointed rod section, one threaded coupling, and the driving bolt. Couplings shall be of a copper alloy. Driving bolts shall be steel with a square end. All threads shall be 5/8-inch-11NC. A ground clamp shall be provided with each assembly to accommodate the connection of a flexible A.W.G. No. 6 ground wire. Two ground wires shall be provided and shall be not less than 15 feet long each.

3.10 Heat exchanger. The heat exchanger shall be oil-fired; shall be either the fire-tube, liquid-tube, or jacketed type; and shall be of all-welded steel construction. The heat exchanger shall be provided with a burner combustion ring of stainless steel or of refractory material having a pyrometric cone equivalent of not less than 32.

3.11 Insulation. The heat exchanger, except the combustion head, oil transmission lines, and overflow lines shall be insulated with either fiberglass felt insulation conforming to MIL-I-22023, type I, class 4, or vermiculite solid

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block insulation for temperatures up to 1800 °F. Fiberglass felt insulation, when used, shall be applied to the outside of the welded steel jacket of the heat exchanger. The insulation shall be covered with a waterproof outer shell of not less than 0.0598-inch-thick sheet steel or not less than 0.090-inch-thick sheet aluminum. Vermiculite solid block insulation, when used, shall be applied to the inside of the welded steel jacket of the heat exchanger. Either insulation shall maintain the temperature of the outer shell and rear head at not more than 125 °F, plus the ambient temperature, after the heater has been operated at rated capacity for not less than 4 hours. If the temperature of the outer shell and rear head is more than 85 °F, plus the ambient temperature after the heater has been operated at rated capacity for not less than 4 hours, the outer shell and rear head shall be shielded or guarded as specified in 3.4.

3.12 Burner. The burner shall be a single burner of the air-atomizing or mechanical-atomizing type and shall burn fuel oil conforming to VV-F-800, grades DF-A, DF-1, and DF-2, or turbine fuel conforming to MIL-T-5624, grades JP-4 and JP-5. The burner shall be equipped with direct electrical ignition and shall be sequenced for automatic operation. The burner turn-down ratio shall be not less than 3 to 1. The combustion controls shall be of the three-position "high-low-off" type or of the modulating type. The low firing rate shall be approximately 33-1/3 percent of the maximum firing rate. The flame-failure reaction time shall be not more than 4 seconds. The burner shall be provided with all pumps, fans, blowers, and other accessories necessary for operation. Two 50-foot, oil-resistant, flexible fuel lines complete with appropriate fittings which can be connected to the fuel pump shall be provided. The fuel pump supply line shall include a duplex-type or dual-type strainer to permit continuous operation while changing one strainer element. Burner components shall be waterproofed.

3.12.1 Operating sequence. The operating sequence shall be as follows:

a. Burner start by means of a manual starting switch.
b. Fan and timer start for four air changes to purge the heater. The total purge time shall be not less than 15 seconds.
c. After completion of purge, when all fuel safety and heater interlocks are closed and the combustion controls are at low fire, the timer shall actuate the igniter and open the main fuel safety shutoff valve.
d. When the flame-sensing device does not sense the main flame within 15 seconds after the main fuel valve opens, the device shall lock the burner out to prevent automatic recycling. When this occurs, a manual restart shall be required.
e. After the main flame is detected, the combustion controls shall release from the low-fire position and go to high-fire position, except on the modulating-type burner which shall adjust to heat demand.
f. After satisfying the heat demand, the combustion controls shall cause the burner to go from high-fire position to the off position and shall return sequentialy to low-fire position and high-fire position upon a new demand for heat, except for the modulating-type burner which shall adjust to heat demand.
g. Flame failure shall shut down the burner automatically and a manual restart shall be required.

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3.13 Controls and safety devices. Controls and safety devices shall include the following:

a. Indicating temperature regulator of the knob type that can be set manually for the desired transfer oil temperature. The regulator shall be adjustable from 100 °F to 500 °F. A dial-type thermometer shall be provided to indicate the actual temperature for comparison with the setting.
b. High-temperature limit control that shall cut off the burner in the event the regulator fails. This control shall be adjustable from 100 °F to 475 °F and shall have a positive cutoff at 475 °F.
c. Low-oil-level, microswitch-type, cutoff control that shall cut off the burner when the oil level in the expansion tank falls below 25 percent of capacity level.
d. Fuel-failure safety shutoff control.
e. Flame-sensing device of the ultra-violet-detector type. The main oil flame shall be sensed as long as it is present.
f. Flame-failure control of the electronic type. This control shall have a reaction time of not more than 4 seconds.
g. Burner-cutoff control in the event of failure of the main circulating pump.
h. All interlocks and accessories to insure sequence of operation as specified in 3.12.1 and to insure operation of all controls and safety devices.
i. Circuit breaker to protect all live circuits, and a means to manually disconnect the main power supply.

3.13.1 Control box. All automatic controls, manual controls, and temperature-checking devices shall be centrally located in a watertight compartment and shall be accessible to an operator while standing on the ground. Controls shall be in accordance with 3.7.

3.14 Transfer oil pumping and piping system.

3.14.1 Main circulating pump. Unless otherwise specified (see 6.2), the main circulating pump shall be of the positive displacement type and shall be mounted either vertically or horizontally. The pump shall be provided with antifriction bearings or high-temperature, bushing-type bearings. The main circulating pump shall be provided with a bypass system and relief valve to handle the fuel pump discharge rate. The bypass system shall maintain fuel tranfer oil flow through the heat exchanger. All bearings shall be greasable and shall be lubricated from an external source. Lubrication of bearings with transfer oil will not be permitted. The main circulating pump shall be positioned so that there shall be a positive head of transfer oil contained in a surge tank located above the suction side of the pump.

3.14.1.1 Oil withdrawal from external user system. The pumping system shall be provided with a means for withdrawing the transfer oil from the external user system for storage within the heater.

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3.14.2 Piping system. The piping system shall provide all necessary valves and fittings to interconnect the heat exchanger, pump, and expansion tank and shall terminate with an inlet and outlet service connection. Service connections shall be 2-inch iron pipe size NPT for size 1 heater, 2-1/2 inch iron pipe size NPT for size 2 heater, 3-inch iron pipe size NPT for size 3 heater, and shall be equipped with caps and cap-retaining chains or terminal valves.

3.14.2.1 Strainer. A full-flow, duplex-type or dual-type strainer shall be incorporated into the inlet line to permit continuous operation during changing of a strainer element. The strainer body shall be steel and the strainer elements shall be corrosion-resistant steel containing not less than 12 percent chromium.

3.14.2.2 Connections. All threaded connections shall be NPT. With the exception of connections that are necessary for removal of components, all threaded connections shall be externally brazed, silver soldered, or welded to prevent leakage. Flanged connections shall be used to facilitate component removal.

3.15 Expansion tank. A steel expansion tank shall be provided. The tank shall consist of a space to compensate for the expansion of transfer oil being heated in the heat exchanger and being circulated to the user system, a space for transfer oil storage, a drain, an overflow, an airvent valve, a provision for filling, a provision for visually determining the oil level, all necessary controls (see 3.13), and all other accessories necessary for heater operation as specified herein. The expansion tank may be used as the cold-oil seal specified in 3.15.1. The expansion tank need not be insulated but shall be isolated from the heat exchanger and located at an elevation above the pump. The arrangement of the expansion tank shall be such that when the heater is operating, the oil in the tank shall not circulate, as the oil level merely rises and falls due to expansion oil. The minimum capacity of the expansion tank shall be twice the volume required by the expansion of the transfer oil in a fully-charged system when the temperature of the transfer oil is raised from 70 °F to 475 °F. Transfer oil in a fully-charged system shall be that amount of oil required to fully charge the heat exchanger and surge tank (see 3.14.1) plus not less than 180 gallons for the external user system. The expansion tank, heat exchanger, and surge tank shall store all the transfer oil.

3.15.1 Cold-oil seal. A cold-oil seal shall be provided to protect the transfer oil against oxidation. A separate uninsulated tank may be used as the cold-oil seal in combination with the expansion tank, and shall be isolated and located as specified in 3.15. The cold-oil seal shall have a vent at the top to permit air to excape and shall incorporate a provision for visually determining the oil level, a drain, and a 50 °F to 500 °F thermometer. After not less than 8 hours of operation of the heater at rated output, the temperature of the transfer oil exposed or in contact with the atmosphere shall not exceed the ambient temperature +75 °F, when measured within 2 inches of the oil surface.

3.16 Electromagnetic interference. The electromagnetic interference emmision characteristics of the heater shall conform to SAE J551.

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3.17 Lubrication. All surfaces that require lubrication shall be provided with a means for lubricating.

3.17.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun confoming to MIL-G-3859 equipped with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where holes are cut in covers, plates, etc., to provide accessibility to fittings, spring-loaded covers shall be provided. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.17.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure lubricating equipment could damage grease seals or other components.

3.17.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, checkplugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drainplug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure when the heater is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided.

3.17.5 Grease lubrication. All grease lubrication, except for the high temperature grease specified in 3.5.1.5, shall be with greases conforming to MIL-G-10924. The heater shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.17.6 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to operating level with winter grades of military oils designated for use in the temperature range of 0 °F to 32 °F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.18 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.18.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 of the South Coast Air Quality Management District.

3.19 Lifting provisions. Permanently affixed provisions that enable the heater to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209, for type II equipment, class 1 or 3 provisions. Lifting provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.20.

3.20 Tiedown provisions. Permanently affixed provisions that permit fastening the heater to the floor or deck of a transportation medium shall be provided. Tiedown provisions shall meet the requirements specified in MIL-STD-209, for type II equipment, class 2 or 3 provisions. All lift/tiedown points shall be labeled "Lift, Tiedown, or Lift-Tiedown", as appropriate in 1 inch high letters in accordance with MIL-STD-129, section 5.

3.21 Identification marking/labels. The heater shall be identified in accordance with MIL-STD-130. All marking/labels shall be applied to the spreader on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.22 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the heater in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

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3.23 Treatment and painting. The portions of the heater normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168. The heater shall then be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns, with the exception that the first article may be camouflage green 383. Heat resistant paint conforming to MIL-P-14105 shall be used on parts of the heater subjected to a temperature of 400° F or higher. Color per MIL-P-14105 shall be in accordance with FED-STD-595, camouflage green 383, and No. 34094 unless otherwise specified.

3.24 Government-loaned property. Unless otherwise specified, the following property as applicable in the quantitites indicated will be loaned by the Government (see 6.5):

Item	Description	Identification	Quantity
1	Pattern drawings	Camouflage patterns	As required

3.25 Data for camouflage patterns. The contractor shall generate camouflage pattern (see 6.2).

3.26 Instruction plates. Each heater shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the heater. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.27 Warning plates. Warning or caution plates conforming to MIL-P-514, type III, composition C, grade A, class 1, shall be provided for all inherent dangers which can stem from unsafe operation or an unsafe operating position. The words "Warning" or "Caution" shall be yellow. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall both be visible and legible.

3.28 Fire extinguisher. The heater shall be provided with a mounting bracket to accommodate a fire extinguisher conforming to A-A-1106, type I, size 10.

3.29 Stenciling. The gross weight of the heater shall be stenciled on each side of the heater. "Lift" or "Tiedown", as applicable, shall be stenciled adjacent to the respective lifting and tiedown attachments. On type II heaters, the T&RA yearbook prescribed tire pressure shall be stenciled TP (specify pounds) on each fender. All stenciling shall be in block-type letters not less than 2 inches high or higher than (Army 4 inches maximum) as specified in MIL-STD-642, 5.1.2. Registration numbers shall be marked on the heater in accordance with MIL-STD-642. Color shall be as specified in MIL-T-704.

3.30 Toolbox. A toolbox shall be provided and shall be of a size to hold maintenance tools and ground rod assemblies (see 3.9.3). The toolbox shall be made of steel 0.0747-inch nominal thickness and shall include provisions for drainage in its mounted position. The toolbox shall have a hinged lid with the

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hinge extending the full length of one edge and shall be provided with not less than two fasteners that will keep the lid closed when the toolbox is subjected to vibration. The toolbox lid shall have an overlapping lip to prevent entry of rain or dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices. The toolbox shall have provisions for locking.

3.31 Type I. Type I heaters shall be powered from an outside source providing 208-volt, 3-phase, 60-cycle, alternating current.

3.31.1 Frame-skid-mounted. The components of the type I heater shall be supported on a welded structural steel framework consisting of angles, channels, or I-beams, reinforced with crossmembers. The framework shall be bolted to a removable skid-type base, or shall be fabricated integral with a skid-type base constructed of standard-size I-beams. The main skid members shall be curved or turned up at both ends for towing, and shall be provided with two towing eyes. The skids and skid fastenings shall withstand towing without damage or permanent deformation when the heater is towed on dry level surfaces at speeds of not less than 2 mph or more than 5 mph.

3.32 Type II. All components of the heater shall be mounted on a pneumatic-tired trailer. The heater configuration shall be either an integral frame type for both heater and vehicular components or a separate frame for the heater installed on a trailer chassis assembly. Except as specified herein, the heater trailer components shall be in accordance with the following, (in order of precedence):

a. Department of Transportation (DOT) Federal Motor Safety Standards (FMVSS).
b. DOT Motor Carrier Safety Regulations (FMCSR).
c. Truck-Trailer Manufacturers Association (TTMA).

3.32.1 Trailer.

3.32.1.1 Weights and dimensions. The gross vehicle weight (GVW) of the heater shall include the weight of the heater components, accessories, attachments, heat transfer oil, lubricants, and vehicular components. Vehicular component ratings shall not be raised to conform to the requirements of this specification. With the heater level on level ground and with the tire inflation pressure as recommended in the T&RA Yearbook and Supplementary Service Data for the imposed GVW loading and highway speed, the physical characteristics of the heater shall be as follows:

a.	GVW, maximum	12,000 pounds.
b.	Vertical lunette load, range	5 to 10 percent of GVW.
c.	Overall length, maximum	240 inches.
d.	Overall width, maximum	96 inches.
e.	Overall height, maximum (in travel condition with stack removed)	102 inches.
f.	Ground clearance, minimum (excluding running gear)	12 inches.

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g.	Lunette height range	31 to 37 inches.
h.	Angle of departure, minimum (SAE J894)	20 degrees.
i.	Suspension limits	
	(1) GVW to 6,500 pounds, maximum	single axle.
	(2) GVW to 12,000 pounds, maximum	tandem axle.

3.32.1.2 Mobility. The heater at GVW shall perform as specified herein when towed over the roads and terrain specified, by a truck matched to the heater GVW. The heater shall operate over smooth, hard-surfaced primary highways at sustained speeds up to 45 miles per hour (mph) and over gravel and unimproved roads at speeds up to 25 mph.

3.32.1.2.1 Trailing and turning ability. The heater shall trail the towing truck without cramping, weaving, side-slipping, or bottoming. The heater shall follow the truck without exceeding the tracking deviation as specified in DOT Motor Carrier Safety Regulation 393.70, when towed over a paved surface. Turns of at least 60 degrees, as measured from straight ahead to the right and to the left shall not cause damage or interference between the heater and the truck.

3.32.1.2.2 Braking. When tested as specified in 4.5.2.2.1, the combined service brakes of the truck and the heater shall control, decelerate, and stop the vechicle combination in accordance with DOT requirements.

3.32.1.3 Intervehicular connections. The heater shall be furnished with an intervehicular electrical harness and two airbrake lines for connecting the towing vehicle electrical and airbrake systems to the heater systems. The airbrake service line shall be located on the curb side and the emergency line shall be on the roadside. All intervehicular connections shall be installed on the heater so that they extend between 36 and 42 inches toward the towing vehicle from the center of the lunette eye. A spring clip shall be installed to hold the electrical harness plug to the drawbar with the harness doubled back on the drawbar. Two fixed dummy couplings shall be installed to hold the airbrake line ends with the lines doubled back. Each dummy coupling shall be mounted perpendicular to the mounting surface and clearance shall be provided for the spring dust cover on the coupling. Intervehicular connections, when connected and when stowed, shall be routed and protected so that dropping of the tailgate of a military truck shall not cause damage. When a 7 and 12 pin receptacle is specified (see 3.32.1.13.1), intervehicular harnesses compatible with each receptacle shall be provided as specified herein.

3.32.1.4 Drawbar components. Each heater shall be equipped with a towing attachment having a lunette conforming to SAE J847 installed in accordance with SAE J849. The lunette shall be supported the entire length of the shank in a housing integral with the towing attachment. The clearance between the shank and the housing shall not exceed 0.035 inch. Upper and lower stops shall hold the square flange of the lunette to prevent rotation. The lunette shall be installed on the towing attachment within 1/8 inch on either side of the centerline of the heater. The towing attachment shall be of the fixed type, constructed and installed to provide a structure necessary to resist pulling,

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bending, and twisting without permanent defromation or damage under all transport conditions specified herein. Two safety chains conforming to SAE J697, with two safety hooks, shall be attached to the trailer frame and extend between 36 and 42 inches forward of the center of the lunette eye.

3.32.1.5 Support legs. Four support legs shall be provided and shall be located at the corners of the heater. The height of the support legs shall be adjustable under load and shall level the heater at any position on a 10-degree slope. The ground clearance under the legs, when retracted or hinged to fold into the travel position, shall not be less than the ground clearance of the first trailer crossmember. The support legs shall be arranged for positive locking into parking and travel positions. A folding-type crank handle for operating the vertical adjustment mechanism shall be provided. The tangential driver's effort at the handgrip to raise the maximum GVW load shall not exceed 30 pounds. The mechanism shall be protected from exposure to foreign matter which would impair functional efficiency. The landing gear shall incorporate a self-aligning land pad which will provide a ground bearing load not exceeding 20 pounds per square inch when the maximum towbar weight specified in 3.32.1.1 is imposed. The edges of the pad shall be turned up.

3.32.1.6 Frame. The frame shall be of sufficient strength and rigidity to support and connect all components, maintain alignment, and distribute the stress of the suspension mounts in order to meet the transport and operating conditions specified herein.

3.32.1.6.1 Rear bumper. A bumper not less than 8 inches in section depth shall be provided at the rear of the heater. The bumper shall extend rearward at least 2 inches beyond any component of the heater. The bumper shall extend the full width of the heater and the ends shall be rounded or wrap-around to the extent necessary to prevent hooking when sideswiped. The bumper shall conform to the safety requirements of DOT FMCSR.

3.32.1.7 Suspension. The heater suspension system shall withstand static and dynamic loads, when operated as specified herein, without permanent deformation, damage, or misalignment. The suspension system, in conjunction with the tires, shall protect the heater from damaging shock and vibration. The suspension system shall have a rated capacity at least equal to the load imposed on each member, measured at the ground. The suspension system shall be of the nonlubricated type. The suspension system shall provide a GVW distribution as specified in 3.32.1.1. The heater suspension shall have sufficient articulation to permit tandem axle heaters to roll over 6-inch-high obstacles with any wheel without loss of tire-to-surface contact of the remaining wheels. Each axle and trunnion mounting shall be cushioned with flexible, elastomer material. Elastic stops or bumpers shall prevent metal-to-metal contact at maximum jounce and rebound position of all wheels. The axle (forward axle or tandem axles) of the heater shall be perpendicular to and symmetrical about the heater's longitudinal centerline. The difference between the diagonal measurements from the center of the lunette eye on the heater centerline to the center end of each front spindle or tandem trunnion shall not exceed 1/16 inch. The alignment of each subsequent axle shall result in these components being parallel to the forward axle spindles or tandem trunnion. The difference between all corresponding spindle

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centers on each side shall not exceed 1/8 inch. Tire and wheel assemblies for all axle combinations shall be balanced within industry limits to eliminate wheel tramp and drumming at road speeds up to 40 mph. On single axle heaters, two shock absorbers shall be provided to control jounce and rebound. The two shock absorbers shall be mounted to the frame and to the axle on each side of the heater with elastomer bushings.

3.32.1.8 Single and tandem axles. The axle(s) shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 150 percent of the heater weight when loaded to maximum GVW. The spindles shall be alloy steel, hardened and ground, and shall be in the correct alignment position with the axle tube. The spindles shall be equipped with tapered roller bearings with cups, bearings, and cones in matched sets from the same bearing manufacturer. The axle hubs shall be of the 6-stud, 8-3/4-inch bolt circle type, conforming to SAE J694.

3.32.1.9 Wheels, rims, and tires. The heater shall be equipped with single demountable disc wheel and tire assemblies. The disc wheels shall be of the heavy duty type with T&RA rim. The rim and wheel disc shall be constructed by the same wheel manufacturer and shall be in accordance with the wheel manufacturer's recommendations for the loads and service specified herein. Rims, tires, inner tubes, and flaps (if furnished), valve cores, and caps shall be in accordance with the T&RA Yearbook. The tires shall be of not less than 100 level quality and shall be of the highway or traction lug type. All tires shall be of the same size and load range. The tires shall be the tube or tubeless type, having a minimum size of 9.00 - 20, load range (12 ply rating). The load imposed on each tire, measured on level ground under loaded travel conditions, shall not exceed the recommended rated loads and inflation pressures of the tires for 50 mph highway speed for continuous highway operation. Valve caps shall be of metal and shall be equipped with elastomer inner seals. Valves shall be of the C-4, heat-resistant types.

3.32.1.9.1 Balancing. Tires, wheels, hubs, and brakedrums shall be inherently in-balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified herein.

3.32.1.10 Fenders and mudguards. The heater shall be equipped with fenders and rear splash guards. The fenders and splash guards shall be in accordance with SAE J321, SAE J682, and SAE J683. The fenders and splash guards shall restrict water and other road surface substances thrown from the heater wheels. The fenders either shall be contoured to the normal axle centers or shall be octagonal shape about the wheels and shall include inner splash shields. The fenders shall support a weight of not less than 200 pounds imposed on any 1-square-foot area at any one time without damage or permanent deformation. The outer edges of the fenders shall be rolled to reduce side road splash. The splash guards shall be of a size not less than 12 by 18 inches, and shall include antisail characteristics. The guards shall be of fiberglass reinforced polyester or of solid polyvinyl chloride homopolymers and monomeric low temperature plasticizers without reinforcement. The physical properties of the guards shall have an ultimate strength of not less than 3,000 psi, a tear strength of not less than 375 pounds per inch, and a flex test life exceeding 1,000,000 cycles.

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3.32.1.11 Brake system. The heater shall be equipped with a waterproof, service and emergency breakaway brake system, actuating foundation brakes on all wheels. The brake system shall comply with DOT requirements as specified in 3.32, except as specified herein, and shall perform as specified in 3.32.1.2.2 under all conditions. All components and lines of the brake system shall be located, installed, and protected to prevent damage or malfunction from water, ground obstructions, and debris and to obtain maximum efficiency and ease of maintenance. All components, with the exception of the foundation brake assemblies, shall be installed forward of the first frame crossmember within the drawbar area. The brake system shall be either straight air type or air-over-hydraulic type, as specified herein. Either brake system shall include (1) service and emergency lines as specified in 3.32.1.11.2; (2) service and emergency air line filters or air cleaners; (3) SAE J10 air reservoir with protected drain cock; and (4) emergency relay valve.

3.32.1.11.1 Air-over-hydraulic brakes. The air reservoir shall have a volume capacity of not less than 900 cubic inches for single-axle heaters and not less than 1750 cubic inches for tandem-axle heaters. One power cluster shall be furnished on single-axle heaters. Two power clusters shall be furnished on tandem-axle heaters. Each power cluster shall be matched to the size of the foundation brakes they operate and shall have their high mortality parts interchangeable with other military-type trailers or semitrailers. On tandem-axle heaters, when furnished, the power clusters shall be arranged to provide either axle-by-axle or brake-by-brake protection. Failure of either hydraulic system shall not prevent actuation of the other system and actuated brakes shall not cause unbalanced braking or swerving of the heater.

3.32.1.11.2 Brake lines. Intervehicular air lines shall be furnished with connectors equipped with metal spring covers, identification tags, and coil spring hose guards on each end. The airbrake lines shall be secured to the heater by cushioned support clamps at intervals not exceeding 24 inches. The brake lines that pass through metal frame members shall be protected by through-frame connectors. The hydraulic flexible hose shall be removable at the axle ends and the foundation brake ends. The brake lines shall be routed from first frame crossmember to the axle assemblies along the interior flanges of the longitudinal frame members on each side. All brake lines shall be installed to facilitate replacement from bulk supply. The brake-line components shall conform to SAE J1402, type D, class II, and SAE J516, 100R5 reuseable fittings.

3.32.1.11.3 Brakedrums. The brakedrums shall be of the antibellmouth type having SAE J431 automotive gray cast iron rubbing surfaces SAE G3500 or SAE G4000. The drums shall be of the heavy-weight type that will provide an adequate heat sink for use with metallic linings.

3.32.1.11.4 Brake linings. The brake linings shall be of the nonfading-from-heat and -from-water-immersion type. The lining shall be of a standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc.

3.32.1.12 Chock blocks and cradles. The heater shall be provided with two chock blocks and two steel cradles. One chock block and cradle shall be installed on each side of the heater so that the block may be positioned either

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to the front or to the rear of the wheel assembly with the captive fastener attached to the heater. The chock blocks shall be as shown in figure 2 and the cradles shall be as shown in figure 3.

3.32.1.13 Vehicular electrical system. Unless otherwise specified herein, the heater shall be equipped with a waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following (in order of precedence):

a. DOT Motor Vehicle Safety Standards (MVSS).
b. DOT Motor Carrier Safety Regulations (MCSR).
c. SAE Handbook, Electrical Equipment and Lighting Section.
d. TTMA Recommended Practices RP No. 9, RP No. 40, RP No. 6, and Bulletin No. 65.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The electrical components shall be in accordance with table I. Item quantitites that are not specified shall be furnished as required (AR) for each heater.

TABLE I. Vehicular electrical system.

Description	Requirements	Quantity Per Heater
7 Contact Receptacle	SAE J 560 and TTMA RP No. 40, 7-wire type with plated metal connectors and waterproof, spring loaded metal cover.	1 ea
Connection box	Waterproof type with either stud and nut terminals or spring loaded terminals.	1 ea min
Circuit breakers	SAE J553 protecting all live circuits.	AR
Wiring	SAE J558 or SAE J878, heavy duty type with 6 rubber or thermosetting insulation only. Wire size SAE J555, min 14 gage.	AR
Wire identification	SAE 2560 or J821	AR
Wiring harness	Polyethylene electrical conduit, each waterproofed with silicone sealant or with one-half overlapping turns of plastic electrical tape.	AR
Grommets	Split type where needed along the wire routing to prevent chafing.	AR

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TABLE I. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per Heater
Supports	Elastomer cushioned support straps or elastomer ties with mounts; wiring support spacing max 36 in span, prevent whipping, chafing, and fatigue damage.	AR
Wiring splices	Mechancial splicing devices; interior waterproofed with silicone lube or compound.	AR
Connectors and terminal	Waterproof types; blade type SAE J858, pin type SAE J928, and spade SAE J561. Blade and pin types fully enclosed and lubricated, prior to assembly with silicone lubricant; spade type equipped with heat shrinkable sleeves and coated with electrical varnish after assembly; all connectors and terminals plated or solder-coated to retard corrosion.	AR
Connector holder clip	Plastic or plastic-insulated metal spring clips; fastened to vehicle; furnished to anchor wire-to-wire friction connectors.	1 ea per friction connector
Running lamps	4 side-marker, 4 clearance, 3 rear identification; per RCCC RP No. S110/1 or RP No. S110/2, flush lamp or low-profile lamp; waterproof and shock resistant types.	11 ea min
Combination lamps	2 turn and tail outboard and 2 stop and tail inboard; Per TTMA RP No. 14 and SAE J759, SAE-IST; flush, elastomer housing with bulb and sealing snap-on, or gasketed lens, or sealed unit with elastomer ring; round 4-in diameter size installed in 4-1/2-in diameter hole; double contact; 3 wire type including ground wire; waterproof and shock resistant type; installed in the rear bumper.	4 ea min

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TABLE I. Vehicular electrical system. (Continued)

Description	Requirements	Quantity Per Heater
Lamp lenses and bulbs	Polycarbonate plastic type lenses; heavy duty or fleet service type bulbs; bulbs mounted in brass vibration-protected sockets with ceramic or plastic insulators and stainless steel or rubber bulb tension devices.	AR
Reflectors	2 on each corner, round type, securely fastened by bolts, screws or rivets.	8 ea min
Wiring diagram	Plate furnished on trailer per TTMA Bulletin No. 65.	1 ea

3.32.1.13.1 Conversion to 24 volt electrical system. Unless otherwise specified (see 6.2), the heater shall be furnished with a 12-contact receptacle conforming to MS75021-1 in addition to the 7-contact receptacle. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage from the military designed towing vehicle. The 24-volt, 12-contact receptacle shall be connected through the necessary resistors into the following 12-volt trailer circuits as follows (letters depict the 24 volt contact terminals and the numbers refer to the SAE J560 receptacle):

A - Do not connect.
B - Connect to circuit No. 3.
C - Do not connect.
D - Connect to circuit No. 1.
E - Connect to circuits No. 2 and 6.
F - Do not connect.
H - Do not connect.
J - Connect to circuit No. 5.
K - Connect to circuit No. 7.
L - Do not connect.
M - Do not connect.
N - Do not connect.

Circuits B and J of the 12-contact receptacle on tactical trucks are combination stop and turn indicator circuits. The normal 12-volt signal lights will function both as turn signals and stop lights and the normal 12-volt stop lights will not be in operation when the trailer is connected to a towing vehicle with a 24-volt power supply. Because of this condition the No. 4 on stop light circuit is not connected to the 24-volt, 12-contact receptacle. Circuit breakers in accordance with SAE J553 shall be furnished. Each receptacle (7- and 12-contact) shall be installed in its own weatherproof connection box, and shall have a spring-loaded cover. Each receptacle shall be permanently identified as to the applicable voltage.

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3.32.2 Power supply. Power to operate the heater shall be supplied by a commercial 15 kW, 60 Hz, Mobile Electric Power (MEP) generator set.

3.32.2.1 Power cable and reel. The main feeder cable assembly shall conform to MIL-C-52286/11 except that the cable shall be type CO-03HLF(3/4-3/10R)1170. A manually-operated, steel reel to contain the 100-foot cable shall be permanently mounted on the heater. Provision shall be made to secure the cable on the reel.

3.32.2.2 Power cable adapter assembly. A power cable adapter assembly shall be provided. The cable assembly shall conform to MIL-C-52286/24 except the length of the cable shall be 5 feet ±6 inches, and the cable shall be type CO-03HLF(3/4-3/10R)1170.

3.32.2.3. Main power supply connectors. One each main power supply connector shall be provided conforming to MIL-C-52286, figures 29 and 43.

3.32.3 Flexible hose. Flexible hose shall conform to MIL-H-18160, class S, 2-inch inside diameter, 8-foot lengths, except protective wire guard and couplings shall not be furnished. The flexible hose shall be capable of hot-oil service at temperatures up to 475 °F. Four 8-foot lengths of the specified flexible hose and four 2-inch, all-steel unions with stainless steel seats shall be provided to enable connection of the flexible hoses to one another and to the inlet and outlet service connections.

3.32.3.1 Container for flexible hose. A storage container shall be provided to accommodate the four 8-foot lengths of flexible hose and four 2-inch steel unions specified in 3.32.3. The container shall be fabricated as specified in 3.30.

3.33 Endurance. When subjected to the tests specified in 4.5.2.6, the heater shall show no evidence of cracking, flaking, shrinking, cement failure, or any other damage.

3.34 Rain. When subjected to the tests specified in 4.5.2.9, the heater shall show no water leakage into the control box, exposed safety components, electrical conduits, or junction boxes.

3.35 Noise limits. The noise produced by the heater shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.2.14. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.36 Workmanship. The heater shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the heater. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and

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sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.36.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping, wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the heater to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the heater shall be at the option of the contractor (see 6.2).

3.36.1.1 Welder certification. Welders shall be certified according to AWS B2.1, Standard for Welding Procedures and Performance Qualifications. Certification shall be made available for review by the contracting officer or designated representative.

3.36.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.36.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

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contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2.1 through 4.5.2.14. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each heater shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each heater shall be tested as specified in 4.5.2.7, except that the heater shall be restarted only one time in lieu of 25 times. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The heater shall be examined for the following defects:

101.	Heater not as specified.	3.1
102.	Safety not as specified.	3.4
103.	Material not as specified.	3.5
104.	Material deterioration and control not as specified.	3.5.1
105.	Dissimilar metals not as specified.	3.5.1.1
106.	Identification of materials and finishes not as specified.	3.5.1.2
107.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the heater.	3.5.1.3
108.	Transfer oil not as specified.	3.5.1.4
109.	Bearings not as specified.	3.5.1.5
110.	Gaskets and seals not as specified.	3.5.1.6
111.	Refractory material not as specified.	3.5.1.7
112.	Structural steel not as specified.	3.5.1.8
113.	Sheet steel not as specified.	3.5.1.9
114.	Sheet aluminum not as specified.	3.5.1.10
115.	Pipe and tubing not as specified.	3.5.1.11
116.	Valves and pipe fittings not as specified.	3.5.1.12
117.	Interchangeability not as specified.	3.6
118.	Common parts not as specified.	3.6.1
119.	Human factors engineering not as specified.	3.7
120.	Circulating pump motor not as specified.	3.9

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121.	Motor starter not as specified.	3.9.1
122.	Internal wiring not as specified.	3.9.2
123.	Ground rods not as specified.	3.9.3
124.	Burner not as specified.	3.12
125.	Transfer oil pumping and piping system not as specified.	3.14
126.	Oil withdrawal system not as specified.	3.14.1.1
127.	Piping system not as specified.	3.14.2
128.	Strainer not as specified.	3.14.2.1
129.	Connections not as specified.	3.14.2.2
130.	Lubrication not as specified.	3.17
131.	Lubricants not as specified.	3.17.1
132.	Lubrication fittings not as specified.	3.17.2
133.	Pressure-release device not as specified.	3.17.3
134.	Filling, draining, and checking provisions not as specified.	3.17.4
135.	Grease lubrication not as specified.	3.17.5
136.	Fungus and moisture resistance not as specified.	3.18
137.	Composition not as specified.	3.18.1
138.	Identification marking/labels not as specified.	3.21
139.	Shipping data plates not as specified.	3.22
140.	Treatment and painting not as specified.	3.23
141.	Paint color not as specified.	3.23
142.	Camouflage pattern data not as specified.	3.25
143.	Camouflage patterns not as specified.	3.25.1
144.	Data for camouflage patterns not as specified.	3.25
145.	Instruction plates not as specified.	3.26
146.	Warning plates not as specified	3.27
147.	Fire extinguisher not as specified.	3.28
148.	Stenciling not as specified.	3.29
149.	Toolbox not as specified.	3.30
150.	Type I heater not as specified.	3.31
151.	Frame-skid-mounted not as specified.	3.31.1
152.	Type II heater not as specified.	3.32
153.	Trailer not as specified.	3.32.1
154.	Weights and dimensions not as specified.	3.32.1.1
155.	Mobility not as specified.	3.32.1.2
156.	Trailing and turning ability not as specified.	3.32.1.2.1
157.	Braking not as specified.	3.32.1.2.2
158.	Intervehicular connection not as specified.	3.32.1.3
159.	Drawbar components not as specified.	3.32.1.4
160.	Support legs not as specified.	3.32.1.5
161.	Frame not as specified.	3.32.1.6
162.	Rear bumper not as specified.	3.32.1.6.1
163.	Suspension not as specified.	3.32.1.7
164.	Single and tandem axles not as specified.	3.32.1.8
165.	Wheels, rims, and tires not as specified.	3.32.1.9
166.	Balancing not as specified.	3.32.1.9.1
167.	Fenders and mudguards not as specified.	3.32.1.10
168.	Brake system not as specified.	3.32.1.11
169.	Air-over-hydraulic brakes not as specified.	3.32.1.11.1
170.	Brake lines not as specified.	3.32.1.11.2
171.	Brakedrums not as specified.	3.32.11.3
172.	Brake linings not as specified.	3.32.1.11.4
173.	Chock blocks and cradles not as specified.	3.32.1.12

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174.	Vehicular electrical system not as specified.	3.32.1.13
175.	Conversion to 24 volt electrical system not as specified.	3.32.1.13.1
176.	Power supply system not as specified.	3.32.2
177.	Power cable and reel not as specified.	3.32.2.1
178.	Power cable adapter assembly not as specified.	3.32.2.2
179.	Main power supply connectors not as specified.	3.32.2.3
180.	Flexible hose not as specified.	3.32.3
181.	Container for flexible hose not as specified.	3.32.3.1
182.	Noise limit not as specified.	3.35
183.	Workmanship not as specified.	3.36
184.	Welding not as specified.	3.36.1
185.	Welder certifications not as specified.	3.36.1.1
186.	Bolted connections not as specified.	3.36.2
187.	Riveted connections not as specified.	3.36.3

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to examination or test, the heater shall be lubricated with oils and greases specified in 3.17 and shall be fully charged with transfer oil as specified in 3.15. Lubricants shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The contractor shall make provisions to dissipate heat output generated by the heater at rated capacity. The heater shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Road.

4.5.2.2.1 Type II. Tow the heater for 25 miles over secondary roads at speeds necessary to average not less than 25 mph. Tow the heater for 25 miles over primary roads at speeds up to 45 mph, to average not less than 40 mph. In addition, tow the heater on smooth paved surfaces to include not less than 15 abrupt stops from a speed of 20 mph in the forward direction, not less than five abrupt stops from a speed of 5 mph in the reverse direction, and not less than 20 sharp turns (10 to the right and 10 to the left) at maximum safe speeds in forward direction. Measure accurately the distance from initial movement of the brake pedal to stopping point for the 15 abrupt stops. The suspension articulation shall be tested by traveling over 6-inch high obstacles. Demonstrate compliance with DOT FMCSR for off-tracking of towed vehicles. Demonstrate compliance of the brake system with DOT requirements specified herein. Noncompliance with any DOT requirement, loss of tire-to-ground contact of remaining wheels while one wheel negotiates 6-inch high obstacles, any operating difficulty; damage to or displacement of any component or accessory installed on the heater; evidence of defective welded, bolted, or riveted connections; or evidence of any leakage from the heater, air lines, and hydraulic lines shall constitute failure of this test.

4.5.2.3 Heat exchanger. Operate the fully-charged heater until discharge flow rate and temperature are stabilized as specified in 3.8. Use a metering device which shall measure the flow of transfer oil within ±2 percent by weight

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of actual flow. Demonstrate the accuracy of the metering device by metering 100 gpm of transfer oil for a period of not less than 5 minutes. The oil shall flow through a closed circuit to the return port on the heater at which point the temperature of the transfer oil should be not more then 360 °F. Determine the pressure drop of transfer oil between inlet and outlet of the heat exchanger. Operate the heater for not less than 1 hour under these conditions and record the burner fuel consumption. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.3.1 Pump. Operate the main circulating pump for 1 hour while pumping transfer oil at a temperature of 450 °F and at a discharge pressure of not less than 50 psig. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.4 Heat exchanger insulation. Operate the fully-charged heater at rated output capacity for not less than 4 hours. Measure the temperature of the outer shell and rear head at not less than four locations on the outer shell and two locations on the rear head. Conduct this test during the last 4 hours of the endurance test specified in 4.5.2.6. A surface temperature at any location on the outer shell or rear head in excess of 125 °F, plus ambient temperature, or shield or guard provisions missing when the surface temperature at any location on the outer shell or rear head is in excess of 85 °F, plus ambient temperature, shall constitute failure of this test.

4.5.2.5 Cold-oil seal temperature. Operate the fully-charged heater at rated output capacity for not less than 8 hours. Measure the temperature of the transfer oil in the cold-oil seal tank or the expansion tank, when used as cold-oil seal, within 2 inches of the oil surface. A temperature at any point in excess of 150 °F, plus the ambient temperature, shall constitute failure of this test.

4.5.2.6 Endurance. Operate the fully-charged heater for not less than 104 hours at rated output capacity of not less than 70 gpm at not less than 450 °F for size 1 heater, 100 gpm at not less than 450 °F for size 2 heater, and 140 gpm at not less than 450 °F for size 3 heater. Operate the heater continuously or for increments of not less than 8 hours. Upon completion of 104 hours of operation, open and examine the heat exchanger. Any evidence of cracking, flaking, shrinking, cement failure, or any other damage to the insulation material or any evidence of leakage, breakage, permanent deformation, or malfuinction shall constitute failure of this test.

4.5.2.7 Cycling. Start the fully-charged heater, operate for 1 hour, and shut down. Restart the heater and run through the complete operating sequence as specified in 3.12.1 for 25 times. Inability of the burner to sequence as specified in 3.12.1 or evidence of leakage or malfucntion shall constitute failure of this test.

4.5.2.8 Expansion tank capacity. Heat the oil in the fully-charged system from 70 °F to 475 °F and measure the amount of oil in the expansion tank. Nonconformance to 3.15 shall constitute failure of this test.

4.5.2.9 Simulated rain. After the performance test has been completed, subject the heater to include the control box, all exposed safety components, electrical conduits, and junction boxes to a simulated rain described as follows

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for a period of 10 minutes: Four sprinkler-type nozzles with a 90 degree conical pattern shall be connected to a water supply system which will provide water to each nozzle at a nozzle pressure of not less than 50 psi. Nozzle capacities at the operating pressure shall be not less than 20 gpm. The four nozzles shall be spaced and directed at each longitudinal side of the heater so that the resulting spray pattern completely encompasses the heater. Any evidence of water inside the control box, exposed safety components, electrical conduits, or junction boxes of the heater shall constitute failure of this test.

4.5.2.10 Lifting provisions.

4.5.2.10.1 Test procedure.

a. Lift the heater and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the heater. Determine the true angle of each sling leg relative to the vertical. Measure the clearance between each sling and the heater. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the heater by anchoring the main frame and subject each slinging provision to a working load of 3.2 times the load it carried when suspended. This force shall be applied in the direction as determined in a. above. An acceptable alternate method of test shall be to lift the heater as in a. above and add weights to the heater until the force on each provision is 3.2 times the initial force. Hold the heater under either of the above loading conditions for not less than 90 seconds.

4.5.2.10.2 Failure criteria. Weld failure, permanent deformation of the heater or any slinging provision, when subjected to the prescribed working load, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.11 Tiedown provisions.

4.5.2.11.1 Test procedure. With the heater anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.5.2.11.2 Failure criteria. Weld failure, permanent deformation of the heater or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.12 Control and safety device. Subject each control and safety device to functional operation for not less than 10 times each. Inability of any control or safety device to function as specified in 3.13 or evidence of leakage shall constitute failure of this test.

4.5.2.13 Electromagnetic interference. The heater shall be tested in accordance with SAE-J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article test report. Failure to meet the requirement of SAE J551 or failure to submit the required reports shall constitute failure of this test.

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4.5.2.14 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) ARC increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Exterior noise levels for drive-by shall not exceed 88dB(A) when tested in accordance with SAE J88. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.14.1 Failure criteria. Failure to reduce the noise level or provide documentation verifying that compliance is either technically or economically unfeasible shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. First article packaging shall be tested for its adequacy in accordance with the rail-impact test as set forth in appendix A to MIL-STD-1186.

4.6.1.2.1 Test failure criteria. Failure of the test shall be based on the failure criteria set forth in appendix A to MIL-STD-1186.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

No.	Defect	A	Comm.
188.	Preservatives and their application not as specified.	5.2.1.1	
189.	Unprotected surfaces not preserved as specified.	5.2.1.2	5.2.2
190.	Trailer components not preserved as specified.	5.2.1.3	
191.	Openings in electric motors not sealed as specified.	5.2.1.4	
192.	Interior surfaces of the burner, fuel pump and lines not coated with preservative as required.	5.2.1.5	
193.	Fuel line opening to fuel tank not sealed as specified.	5.2.1.5	
194.	Power cable not contained on reel as specified.	5.2.1.6	

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195.	Exposed cable ends not sealed as specified.	5.2.1.6	
196.	Flexible hose not positioned in container provided.	5.2.1.7	
197.	Hose container lid not secured as specified.	5.2.1.7	
198.	Repair parts not preserved as specified.	5.2.1.8	
199.	Maintenance tools not preserved as specified.	5.2.1.9	
101a.	Technical publications not preserved as specified.	5.2.1.10	
102a.	Exposed chains, when utilized, not preserved as specified.	5.2.1.11.1	
103a.	Drivebelts and pulleys, when utilized, not preserved as specified.	5.2.1.11.2	
104a.	Enclosed gears and chains, when utilized, not preserved as specified.	5.2.1.11.3	
105a.	Flame safeguard relay, when utilized, not preserved as specified.	5.2.1.11.4	
106a.	Consolidation not as specified.	5.2.1.12	
107a.	Commercial preservation not as specified.		5.2.2
108a.	Skid-mounted heaters not packed as specified.	5.3.1.1	5.3.2
109a.	Trailer-mounted heater not prepared for shipment as specified.	5.3.1.2	5.3.2
110a.	Marking not as specified.	5.4.1	5.4.2

5. PACKAGING

5.1 First article pack. When specified (see 6.2), the contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may be prepared utilizing either the first article heater or a production heater. When the first article model heater is utilized, the preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article model heater and production heaters.

5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Unprotected surfaces. Unpainted exterior metal surfaces of components not specifically specified herein and requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.3 Trailer. Components of the trailer shall be preserved in accordance with the level A (mobile) preservation and packaging requirements of MIL-STD-281.

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5.2.1.4 Electric motors. Openings into electric motors shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.5 Burner. Interior surfaces of the burner, fuel pump, and lines shall be coated with P-10 preservative, type I, grade 10. The fuel line opening to the fuel tank shall be sealed with a plastic or metal cap or plug, as applicable.

5.2.1.6 Power cable. The cable shall be contained on the reel and secured to prevent unreeling. The exposed end of the cable shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.7 Flexible hose. The flexible hose shall be placed in the container. The container lid shall be secured in a manner to prevent pilferage of contents.

5.2.1.8 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A.

5.2.1.10 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.11 Design features. When the following components are utilized in the design features of the burner, they shall be preserved as specified herein.

5.2.1.11.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Unpainted sprockets shall be coated with type P-1 preservative.

5.2.1.11.2 Drivebelts and pulleys. Belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.11.3 Enclosed gears and chains. Gears and chains enclosed within housings and with the housing filled with oil to the operating level as specified in 3.17.6 shall be actuated to assure complete coverage with the lubricant.

5.2.1.11.4 Flame safeguard relay. The flame safeguard relay shall be removed from the control box and preserved in accordance with MIL-P-116, method IC-1. The preserved relay shall be secured within the control box. A tag conforming to UU-T-81, type A, shall be attached to the control box indicating "Flame safeguard relay is disconnected and is secured within. Connect relay before operating heater."

5.2.1.12 Consolidation. The repair parts, maintenance tools, technical publications, ground rod assemblies, removed drivebelts when required, and loose components shall be consolidated in the toolbox when space permits. When space does not permit, the remainder of the items that will not fit in the toolbox

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shall be consolidated in a close-fitting box conforming to PPP-B-601, domestic type, style optional. The box shall be secured to the heater with strapping. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Strapping shall be finish B. The box shall be secured in a manner that will not increase cubage or interfere with lifting or towing of the heater. The toolbox lid shall be secured to prevent pilferage.

5.2.2 Commercial. Each complete burner shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Type I - skid-mounted heaters. Each skid-mounted heater, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-3774, type I, style A. Loading, bracing, blocking, bolting, cushioning and closure shall be in accordance with the appendix to the crate specification and with MIL-STD-1186.

5.3.1.2 Type II - trailer-mounted heaters. Each trailer-mounted heater, preserved as specified in 5.2, shall be prepared for unboxed (mobile) shipment. The securing of the consolidation container to the trailer and the securing of the toolbox lid shall be as specified in 5.2.1.12. Wheels shall be free to rotate.

5.3.2 Commercial. Type I and type II heaters, preserved as specified in 5.2, shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Level A. In addition to special or identification markings required by the contract or order, the marking of the heaters shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each heater shall be marked to show the cube and gross weight.

6. NOTES

6.1 Intended use. The heaters are intended to heat and circulate transfer oil for use with asphalt plants, asphalt melters, or other similar heating applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).
c. Time frame required for submission of first article model, and number of heaters required (see 3.2).
d. When the identifiction of materials and finishes are required (see 3.5.1.2).

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e. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When transfer oil is not to be furnished (see 3.5.1.4).
g. When pipe and tubing will be other than specified (see 3.5.1.11).
h. When valves and pipe fittings will be other than as specified (see 3.5.1.12).
i. When the circulating pump motor will be design C (see 3.9).
j. When main circulating pump shall be other than as specified (see 3.14.1).
k. When a 12-contact receptacle is not required (see 3.32.1.13.1).
l. Color when other than specified (see 3.23).
m. When Government-loaned property is not required (see 3.24).
n. Time frame when camouflage pattern data is required (see 3.25).
o. When the welding process shall not be at the option of the contractor (see 3.36.1).
p. When a first article pack is required, and time frame required for submission of first article pack (see 5.1).
q. Degree of preservation and degree of packing required (see 5.2 and 5.3).
r. Any special markings (see 5.4.1).

6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract data requirements list (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410.6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements	Applicable DID NO.
a. 3.25	Color photograph prints	DI-MISC-80192
b. 3.25	Camouflage line art data	DI-MISC-80176

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one heater. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article documents.

6.4 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubri-

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cants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research & Development Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.24.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the heaters in accordance with this specification.

6.7 Packing. Degrees of packing have not been specified as the packing specified is applicable for all degrees of packing. For purposes of preservation/packing level marking, the marking shall be level A.

6.8 Subject term (Key Word) listing.

Heaters, asphalt melters
Heaters, asphalt plant
Heaters, hot oil, trailer mounted, skid mounted

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - AT

Project 3895-A226

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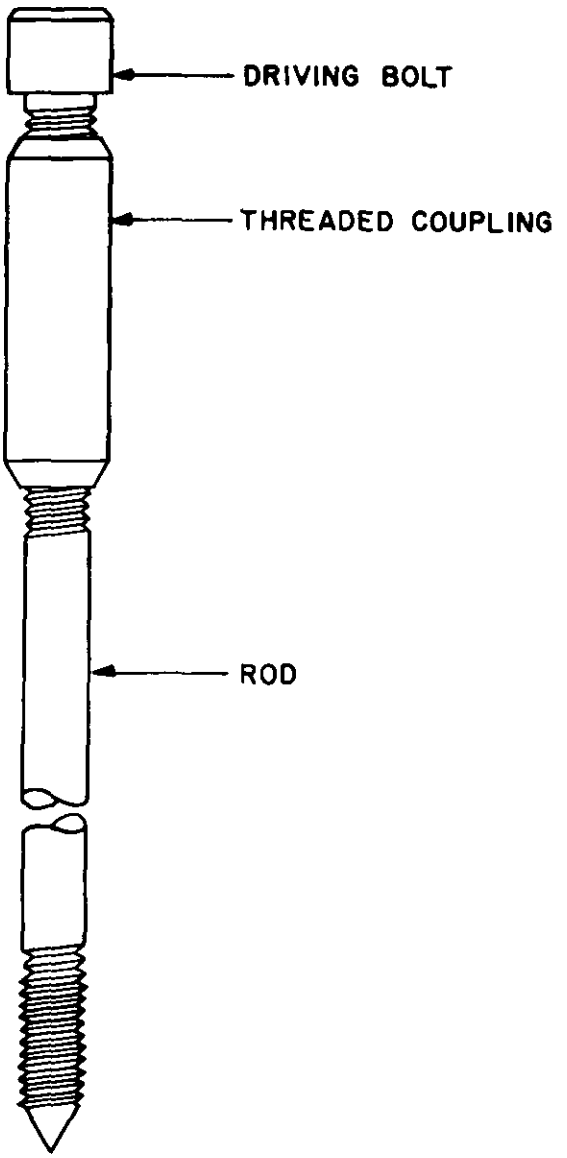
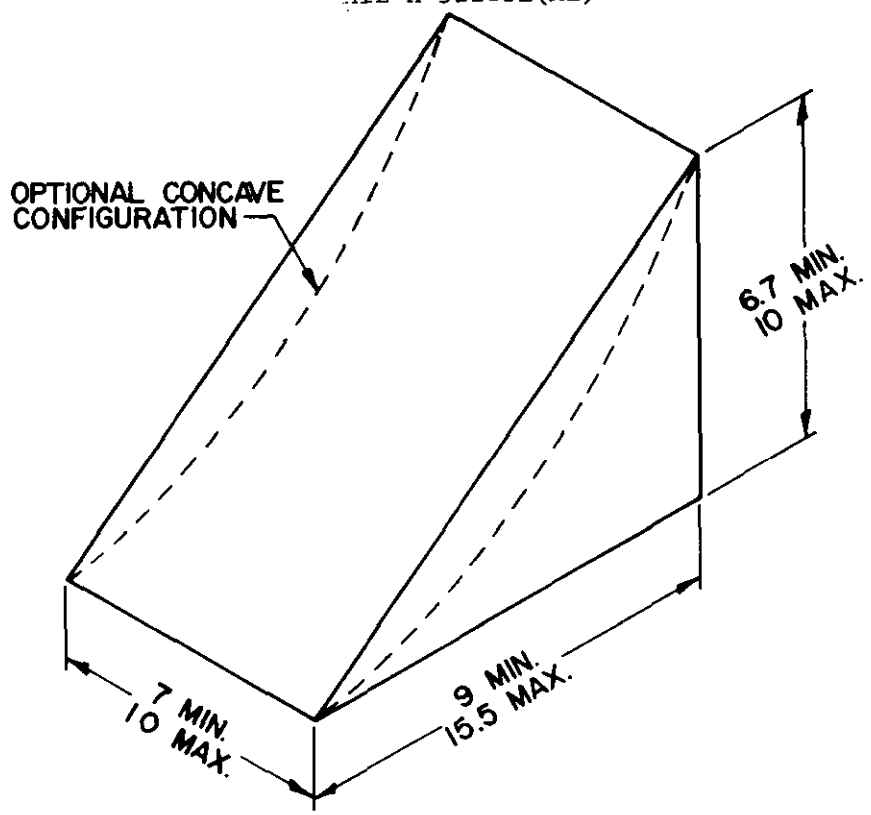


FIGURE 1. Rod assembly, ground.

X-4477

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CHOCK MATERIAL NOTE

1. ALUMINIUM
2. SEMI-RIGID ELASTOMER
3. MAXIMUM CHOCK WEIGHT - 11.5 LB.

CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 2. Chock block assembly.

X 603 B

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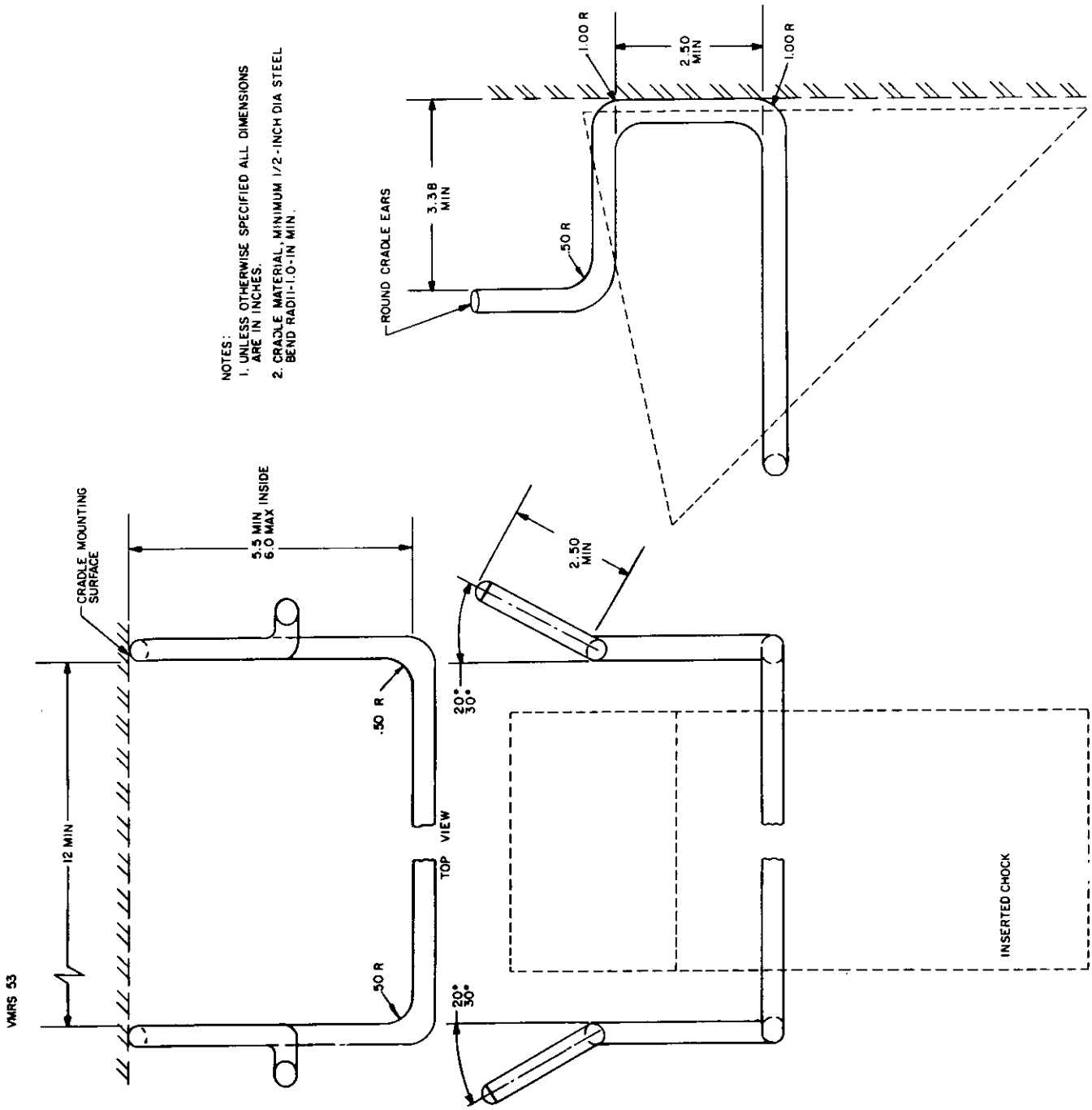


FIGURE 3, Cradle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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